

NeuroXAI - An Explainable AI Framework for Parkinson's Disease Risk Assessment

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Abstract – Parkinson's Disease is a chronic neurodegenerative disorder causing gradual deterioration in motor skills and neurological stability, thus requiring early detection and subsequent monitoring to be intervened with the appropriate care. However, conventional diagnosis requires clinical expertise and is not always consistent in the early stages. This has accelerated the need for AI-based healthcare systems for automated disease prediction. Most of the current work is either based on medical images or clinical data separately, which limits their holistic approach to a reliable diagnosis. NeuroXAI is a novel multimodal framework developed to overcome such limitations. It integrates clinical data analysis with brain MRI-based prediction using explainable artificial intelligence techniques. The system applies various machine learning models-SVM, Random Forest, XGBoost, and Logistic Regression-to clinical risk and deep learning architectures CNN, ResNet, MobileNet, and EfficientNet-for image-based analysis. Such a framework enables model comparison and selection of the best model based on metrics such as accuracy, precision, and recall. The other essential features include multimodal prediction, explainability through SHAP and LIME for clinical data, Grad-CAM, and Integrated Gradients for visual interpretation of MRI images. Moreover, the system develops AI-driven clinical reports with large language models that give human-readable insight and recommendations. The experiment shows promising results with regards to high prediction accuracies and consistency. Particularly, XGBoost achieved 94.3% accuracy in predicting clinical data, whereas MobileNet predicted MRI images with 98.8% accuracy.. This will eliminate most existing concerns about interpretability, dependency on single-modal systems, and limited clinical trust. NeuroXAI is therefore a scalable and effective solution for real-world detection and decision support for Parkinson's Disease through its multiple models, explainability techniques, and user-friendly interface without additional hardware..

Keywords— Parkinson's Disease, Explainable AI, Multimodal Learning, Machine Learning, Deep Learning, SHAP, Grad-CAM, Medical Image Analysis.

I. INTRODUCTION

Parkinson's disease (PD) is a progressive neurological disorder that affects movement, balance, and coordination and seriously deteriorates the quality of life for most patients. Because of the increased prevalence of neurological disorders, there is an enhanced need for early diagnosis and continuous monitoring for timely intervention and effective treatment. Usually, the traditional diagnostic methods are clinical and subjective; hence, they are prone to error and inconsistency, especially in the early stages of the disease.

Over the past decade, Artificial Intelligence (AI) has been one of the emerging potent technologies in healthcare, leading to automated disease prediction from clinical data and medical images. Different AI-based approaches have been

proposed for detecting Parkinson's disease (PD) from clinical symptom analysis and brain imaging techniques. While most of these methods succeed in most cases, they are generally independent and do not offer a holistic view of the disease. Most existing models are black boxes, and the output is given without any explanation, thereby limiting their clinical application.

In addition to improving prediction accuracy, the integration of multimodal data enables the system to capture both symptomatic and structural changes associated with Parkinson's Disease. Clinical data reflects functional impairments, while MRI images capture underlying neurological alterations. Combining these complementary sources increases the prediction system's robustness and minimizes the chance of misclassification that a single modality might yield. This holistic perspective is crucial for early-stage diagnosis where symptoms may be subtle and difficult to interpret.

Some systems are equipped with explainability techniques; however, they are usually unimodal and hence do not give a complete view of the diagnosis. Most solutions also lack a provision to compare different models; thus, determining which one can best predict the outcome becomes difficult. Therefore, there is a gap in developing a unified, interpretable, and flexible system for diagnosis using clinical and imaging data.

NeuroXAI, the proposed system, builds a multimodal explainable AI framework for PD detection. Clinical data will be analyzed with machine learning models Support Vector Machine (SVM), Random Forest, XGBoost, and Logistic Regression, while medical images will be analyzed with CNN, ResNet, MobileNet, and EfficientNet deep learning architectures.

The system further incorporates sophisticated explainability techniques, which contribute to clarity and trust. SHAP (SHapley Additive Explanations) and LIME (Local Interpretable Model-agnostic Explanations) explain the contribution of clinical features to the outcome. In contrast, Grad-CAM (Gradient-weighted Class Activation Mapping) and Integrated Gradients explain the contribution of images to predictions. It provides a structure for model evaluation and

comparison, thus selecting the best model based on performance measures such as accuracy, precision, and recall. The primary proposed work components are multi-way disease prediction, model comparison, dual explainability, and AI-generated clinical insights. Thus, this approach provides a secure, precise, and reliable diagnosis by integrating data from different sources and methodologies. Therefore, this will ease physicians' decision-making and enhance the applicability of AI-powered solutions in real-life health care scenarios.

II. RELATED WORK

Here, we review the existing literature on systems that detect Parkinson's Disease based on AI, their methodologies, limitations, and the motivation for a multimodal explainable framework.

The use of machine learning techniques for Parkinson's Disease prediction using clinical data has been explored in several studies. S. Prashanth et al. [1] used Support Vector Machine (SVM) and decision tree based models for classification based on biomedical voice measurements and clinical features. The method was accurate; however, it was limited to handcrafted features and was thus difficult to interpret, which made it difficult to translate into clinical practice.

A. Tsanas et al. [2] proposed regression-based and machine learning models to analyze speech signals for detecting Parkinson's Disease. Although promising predictive performance was demonstrated, this was limited to a single data modality and had little explainability to justify the model decisions..

S. Pereira et al. [3] proposed using Convolutional Neural Networks (CNNs) in medical imaging for classifying cerebral images. Although the model could identify spatial characteristics, it worked as a black-box system and did not disclose the reasons behind its prediction..

In this regard, K. He et al. [4] proposed deep residual networks (ResNet) to facilitate the extraction and better classification of features from images.

R. Selvaraju et al. [5] in "Grad-CAM: Visual Explanations from Deep Networks via Gradient-based Localization" proposed an image visualization technique, Grad-CAM, for interpreting image classifiers' outputs by localizing critical regions of an image responsible for classification via gradients. Similarly, SHAP and LIME are widely used techniques for interpreting machine learning models that allow researchers to quantify the contribution of each feature to their prediction [6]. However, thus far, these two techniques have been predominantly studied individually without being integrated into an overarching unified diagnostic strategy..

Multimodal (MM) neuroimaging studies have started to use both clinical and imaging data to enhance the accuracy of their

MM predictive models. Unfortunately, the majority of current MM studies are hampered by a lack of integration between the two data sources, limited or no comparison of model accuracies, and limited interpretability – all of these issues significantly reduce their potential for clinical translation..

At this time, the evidence shows that Artificial Intelligence has great potential for assisting in the diagnosis of Parkinson's Disease; however, a number of barriers impede the successful implementation of AI in this area, particularly with respect to the MM integration aspect.

Although prior research demonstrates the effectiveness of individual machine learning and deep learning approaches, there remains a lack of unified frameworks that simultaneously address multimodal integration, model comparison, and interpretability. Most studies focus either on improving accuracy or adding explainability, but rarely both in a single system. This gap calls for an integrated approach like NeuroXAI, which ensures high performance, transparency, and usability in real-world clinical settings.

III. METHODOLOGY

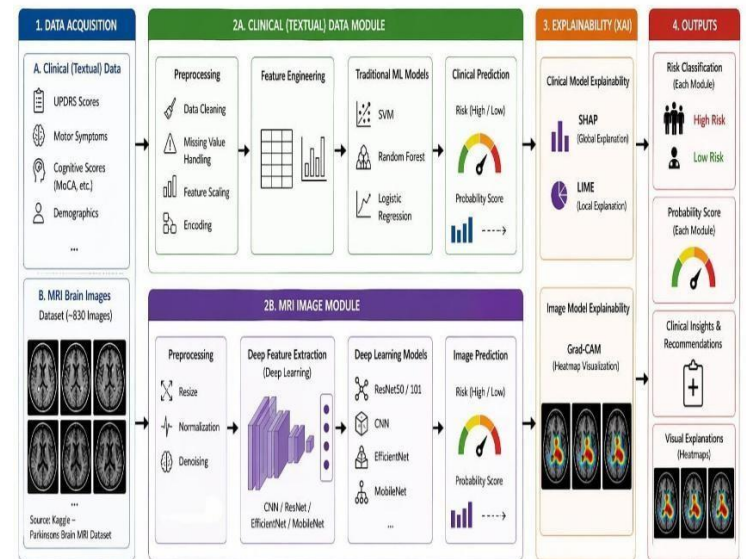


Figure 1

Figure 1 shows the general architecture of NeuroXAI, combining clinical data analysis with medical image-based prediction in detecting Parkinson's disease. Model training, evaluation, and real-time inference are possible, along with explainability mechanisms to foster transparency. The detailed workflow is explained below :

A. Data Collection

In this work, the data were bimodal:

Clinical Data: Features such as tremor, rigidity, bradykinesia, postural instability, and cognitive assessment scores have been widely used to diagnose Parkinson's disease.

TABLE I

CLINICAL DATASET

Characteristics	Normal	Parkinson's	Total
Number of Records	801	1304	2105
Training Split	561	913	1474
Test Split	240	391	631

Medical Imaging Data: Brain MRI scans of PD and healthy control subjects.

TABLE II

MRI DATASET

Characteristics	Normal	Parkinson's	Total
Number of images	610	221	831
Training Split	488	177	665
Validation Split	61	22	83
Test Split	61	22	83

The imbalance problem occurred in the case of MRI image data sets, which had 610 images that were normal and 221 images that were related to Parkinson's disease. To reduce the bias that may occur due to the presence of more normal images, the technique of class weightage was employed. This means that the weights assigned to the minority classes were much higher than those for the majority class.

The dataset has three subsets for model development :

- Training set: Pattern recognition and feature extraction by the model.
- The validation set: For getting optimal Hyperparameters and early stopping so as to prevent model from overfitting.
- The test set: Performance evaluation of models on unseen data.

To ensure consistency in data, preprocessing pipelines were designed separately for clinical and imaging modalities while aligning the labeling and class distribution. This will reduce the bias and ensure that the two modalities contribute equally during the learning process.

Therefore, such a multimodal dataset reflects the patient's temporal evolution, and generalization of the proposed system is more feasible.

B. Processing clinical data and adding new features

Before using the clinical data, the following preprocessing steps are performed:

- Handling missing values
- Normalizing the features
- Encoding categorical inputs (if any)

The derived features are fed into several machine learning models. They describe the severity and progression of Parkinson's symptoms and thus form an essential basis for determining risk levels.

C. Machine Learning Models for Medical Predictions

Various machine learning classification algorithms are thus implemented:

- Support Vector Machine (SVM)
- Random Forest (RF)
- XGBoost
- Logistic Regression (LR)

Performance tuning through hyperparameter optimization was done through grid search and cross-validation. This ensures that all models run under their best configurations and hence will be robust and less prone to overfitting.

TABLE III

OPTIMAL HYPERPARAMETER CONFIGURATION OF MODELS

MODEL	Key Hyperparameters
Logistic Regression	Solver=lbfgs,max_iter1000
SVM	kernel = rbf, C = 1.0, gamma = scale
Random Forest	n_estimators = 200, random_state = 42
XGBoost	n_estimators=200,learning_rate=0.05, random_state=42

D. Image-Based Feature Extraction using Deep Learning

Deep learning architectures for medical images include CNN, ResNet, MobileNet, and EfficientNet. All MRI images will have to be resized and normalized before input into any model.

Convolutional layers will take the input images and convert them into a hierarchy of spatial features. Ultimately, this hierarchy helps identify significant features in the spatial structure related to Parkinson's disease. Then, the previously extracted features will be classified on the fully connected layers.

Combined global and local explainability techniques yield interpretable overall model behavior and individual predictions. This two-level explanation is essential in medicine, where trust in the model's decision and its validation are paramount.

The proposed system includes explainability technologies for improving the understanding of both modalities.

Explainable AI Integration

Clinical Data Explainability: SHAP and LIME explain how much each feature interacts with the prediction.

In this way, combining global and local explainability techniques will result in overall model behavior and individual predictions being interpretable. Such a two-level explanation is especially relevant in medical applications, where trust and validation of model decisions are involved.

Image Explainability: Grad-CAM and Integrated Gradients methods explain features' interaction with prediction.

They are transparent methods that explain the basis for prediction.

E. Model Evaluation and Comparison

All models are evaluated using the following performance metrics::

- Accuracy
- Precision
- Recall

A comparative study will follow the best model to determine its performance. In this way, the system will be able to choose the most accurate model for prediction according to the data characteristics.

In addition to quantitative metrics, qualitative analysis was also performed using visualization techniques such as confusion matrices and ROC curves to better understand model behavior across different classes.

F. Output Generation and Clinical Interpretation

During inference, the system produces:

- Predicted class (Parkinson's/Normal)
- Confidence score
- Feature importance (clinical data)
- Heatmap visualization (images)

Furthermore, it produces AI-based clinical insight and recommendations to guide clinicians in making choices.

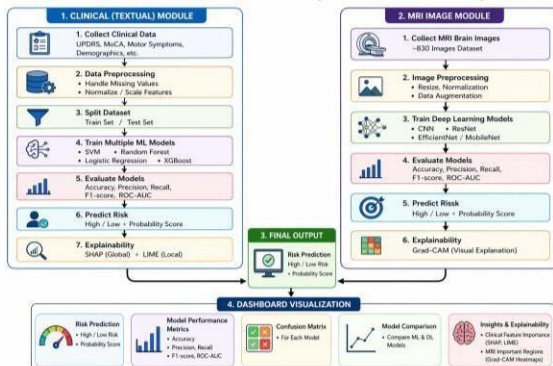


Figure 2

Figure 2 presents a dual-module system combining clinical (textual) data and MRI images for Parkinson's disease risk prediction. The clinical module processes patient data and applies machine learning models (SVM, Random Forest, Logistic Regression, XGBoost), while the MRI module uses deep learning models (CNN, ResNet, EfficientNet, MobileNet) on preprocessed brain images. Both modules output risk predictions (high/low) with probability scores and use explainability methods (SHAP, LIME, Grad-CAM). The final output is result integration, displayed on a dashboard with performance metrics, confusion matrices, model comparison, and interpretability insights.

IV. EXPERIMENTAL RESULTS AND DISCUSSION

The proposed NeuroXAI system's performance was evaluated for clinical (textual) and image-based modules. The experimental analysis focuses on model performance, comparative evaluation, and result interpretability.

A. Clinical Model Performance

The clinical data module was tested on several machine learning models: SVM, Random Forest, XGBoost, and Logistic Regression. The training process showed stable learning, where the models attained high accuracy with close performance across validation and test datasets.

TABLE IV

PERFORMANCE COMPARISON OF MACHINE LEARNING MODELS

Model	Accuracy	Precision	Recall
Logistic Regression	80.0	82.7	85.8
SVM	83.4	84.0	90.4
Random Forest	92.9	93.9	94.6
XGBoost	94.3	95.8	95.0

Of the models tested, XGBoost outperformed other models in terms of accuracy and generalization; hence, it is the best model for predicting clinical risk. The model comparison framework allows a user to analyze and choose the best-performing model dynamically.

The consistent performance across different models indicates that the selected features are highly relevant for Parkinson's disease prediction. It further substantiates the effect of preprocessing and feature engineering.

B. Image-Based Model Performance.

The image module is further tested against various deep learning architectures, including CNN, ResNet, MobileNet, and EfficientNet. The training has been well converged, as seen by increasing accuracy and decreasing loss values over the epochs.

TABLE V

PERFORMANCE COMPARISON OF DEEP LEARNING MRI MODELS

Model	Accuracy	Precision	Recall
Custom CNN	95.2	95.8	94.3
ResNet	83.7	84.5	82.1
MobileNet	98.8	99.1	98.2
EfficientNet	73.5	74.2	72.0

The best-performing model MobileNet yields an extremely high classification accuracy, thus implying that complex patterns within brain MRI images have been learned. This further indicates that the model is robust concerning image quality, intensity, and background noise changes.

Data augmentation techniques therefore contribute to better generalization by exposing the models to variations in orientation and image intensity, which reduces overfitting.

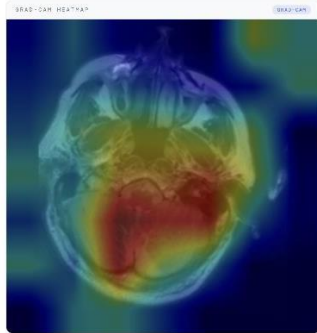
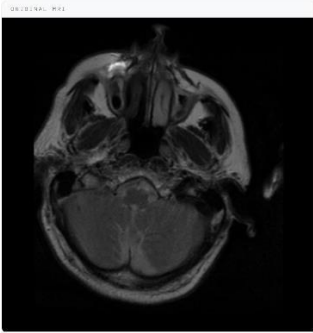
C. Explainability Analysis

Another key advantage of the proposed system is the embedding of explainable AI techniques in both modules:

Clinical Module: SHAP and LIME determine the contribution of each feature (e.g., tremor, rigidity) to the prediction. Thus, these explainability methods allow straightforward insight into how symptoms influence risk assessment.

Raw MRI

GRAD-CAM



RAW MRI

Integrated Gradients

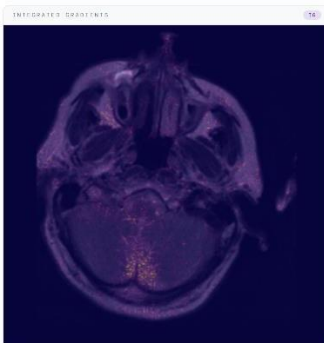


Image Module: Grad-CAM and Integrated Gradients produce heatmaps of salient regions in MRI images. Such salient regions are consistent with medically relevant areas, thus indicating that the model is paying attention to essential features rather than noise.

These outputs from explainability will be able to substantiate the predictions made by the model and provide insights that are clinically valid and, hence, assist physicians in understanding the patterns of diseases and making decisions accordingly.

Thus, explanations at the feature and pixel levels complement each other to make the system transparent and trustworthy.

D. System Interface and Output Analysis

It is an interactive and user-friendly interface for prediction and analysis.

It displays:

- Selected model and prediction output
- Confidence score
- Feature importance (clinical data)
- Heatmap visualization (image data)

Moreover, the system outputs AI-based clinical insight: risk level, disease stage, and suggestions for further medical examination, thereby improving usability and assisting in decision-making by medical staff and users.

E. Robustness and Generalization

The proposed system shows high generalization over both modalities. Clinical models are robust to different feature distribution modalities, and image models remain robust over changes in image conditions.

Thus, combining different models and explainability techniques makes the prediction accurate, interpretable, and reliable.

F. Discussion

These results show that the NeuroXAI framework overcomes some of the critical shortcomings of existing systems. Unlike conventional single-modality approaches, the proposed system combines clinical and imaging data, thus achieving a more holistic diagnosis. Model comparison helps choose the best model, while explainability techniques enhance trust and transparency. However, system performance depends on the quality and variability of the datasets; hence, larger and more varied datasets would further improve it. In terms of computational efficiency, the NeuroXAI framework can be deployed for implementation across many clinical settings. The prediction model, which applies machine learning approaches such as XGBoost and Random Forest, is computationally efficient and can therefore be implemented using general-purpose computing devices that are available in hospitals. While the application of deep learning algorithms to predict MRI images needs greater computational power and thus GPU usage during model training, the application of light models like MobileNet makes it possible to predict MRI images in real-time using a hospital computer without the need for additional hardware.

Furthermore, the modular design of the NeuroXAI framework allows flexibility in extending the system with additional data modalities such as speech signals or wearable sensor data, making it adaptable for broader neurological disorder analysis.

V. CONCLUSION AND FUTURE SCOPE

NeuroXAI: A Multimodal Explainable AI Framework for Parkinson's Disease Detection integrates clinical data analysis and medical image-based prediction into a holistic and reliable diagnostic system. The framework uses multiple machine learning models, such as Support Vector Machine (SVM), Random Forest, XGBoost, and Logistic Regression, for clinical risk assessment and several deep learning architectures, such as CNN, ResNet, MobileNet, and EfficientNet, for image-based analysis. The system, therefore, builds transparency and trust in AI-led medical decisions by incorporating explainability techniques SHAP and LIME for clinical data and Grad-CAM and Integrated Gradients for interpreting images..

This enables model comparison, the selection of the best model according to the performance metrics, and, hence, the best prediction accuracy. Moreover, the system outputs are easy to interpret, such as confidence scores, feature importance, heatmap visualization, and clinical insight from AI, which fits into the practical use of the system within healthcare. Experimental results show that the proposed system has reasonable predictive performance and modality-wise uniformity in results, overcoming significant limitations of traditional single-modality and black-box approaches .

However, some challenges remain: dependence on dataset quality, computational requirements for deep learning models, and the need for datasets that is large and diverse to improve generalization to different populations. Moreover, real-time deployment in resource-constrained environments may require additional optimization.

Multimodal fusion techniques could be included to appropriate fusion between clinical and imaging data for better prediction. The most recent transformer-based architectures and lightweight models reduce computational complexity and enhance efficiency. Further, the system can be enhanced with longitudinal patient data, real-time monitoring, and integration into electronic health record (EHR) systems. Federated learning approaches could also be considered to maintain data privacy while allowing for joint model improvement. In this sense, NeuroXAI could become a robust and scalable clinical decision support system contributing to early diagnosis and management of Parkinson's Disease.

Explainability, combined with multimodal learning, thus not only increases diagnostic performance but also fills the gap between AI systems and clinical adoption due to transparent and interpretable predictions.

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